VPN Note

GRE Tunnel (Generic Routing Encapsulation) A tunneling protocol that encapsulates an inner packet inside an outer IP packet to create a logical point-to-point link between two routers. It treats the underlying network as a transit medium but does not inherently secure the traffic.

- **Advantages:** It natively supports multicast and broadcast traffic, making it capable of running dynamic routing protocols like OSPF. It also provides a logical interface (Tunnel0) that can be monitored and routed just like a physical interface.

- **Disadvantages:** It provides zero confidentiality or peer authentication; anyone capturing packets can read the inner traffic in plaintext. It also adds encapsulation overhead, requiring adjustments to MTU and TCP MSS.

Policy-Based IPsec VPN A site-to-site VPN that uses military-grade encryption and cryptographic hashes to secure data traveling over a public network. Rather than using a virtual tunnel interface, it relies on an Access Control List (ACL) to identify "interesting traffic" and encrypts only the packets that match that policy.

- **Advantages:** It provides strong data confidentiality, peer authentication, and data integrity. It is highly secure and universally supported across nearly all enterprise networking equipment.

- **Disadvantages:** It strictly supports unicast IP traffic and cannot encapsulate multicast or broadcast packets natively. Because of this, you cannot run dynamic routing protocols directly over a standard IPsec VPN.

GRE over IPsec VPN A hybrid design that encapsulates LAN traffic inside a GRE tunnel first, and then applies an IPsec protection policy (often using transport mode) to encrypt the GRE packets as they leave the WAN interface.

- **Advantages:** It delivers the best of both protocols: the multicast routing and protocol flexibility of GRE, combined with the robust security and encryption of IPsec. This design is explicitly used when a network requires dynamic routing protocols over a secure VPN.

- **Disadvantages:** It introduces double encapsulation overhead (both GRE and IPsec headers), making MTU and TCP MSS tuning critical to prevent packet fragmentation and dropping. The configuration is also significantly more complex to deploy and troubleshoot.

IPsec Phase-by-Phase Breakdown

The IPsec negotiation process, which applies to both Policy-Based IPsec and GRE over IPsec, occurs in two distinct phases:

Phase 1: IKE (ISAKMP) Negotiation

The routers authenticate each other and establish a secure, encrypted management channel. They agree on a specific set of security parameters to protect subsequent negotiations.

- **Authentication:** Verifying the peer's identity (e.g., using a pre-shared key).

- **Encryption & Hash:** Selecting the algorithms to secure the Phase 1 channel (e.g., AES-256 and SHA).

- **Key Exchange:** Using a Diffie-Hellman group (e.g., Group 14) to securely exchange cryptographic keys over the untrusted network.

- **Verification:** Confirmed by running show crypto isakmp sa, which should display an established state.

Phase 2: IPsec Security Association (SA) Establishment

Once the secure Phase 1 channel is active, the routers negotiate the parameters for encrypting the actual user data.

- **Transform Set:** The peers agree on the encryption and hashing algorithms (e.g., esp-aes 256 and esp-sha-hmac) applied directly to the data payload.

- **Traffic Selectors:** The routers define exactly what traffic gets encrypted using matching Access Control Lists. In a policy-based VPN, this is the LAN-to-LAN subnets. In a GRE over IPsec VPN, the protected traffic is the GRE protocol (IP protocol 47).

- **Verification:** Confirmed by running show crypto ipsec sa, where the encapsulation and decapsulation counters will increase as data successfully passes through the VPN.